

IPhO Style Multi Problem Answers

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·1· Answer

Resolve gravity parallel to the incline and apply Newton second law.

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2. Answer

Use angular frequency $\omega = \sqrt{k / m}$.

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(3) Answer

Compute isothermal work and use adiabatic relation.

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Problem 4 Answer

Motional emf equals $B L v$ with Lenz law direction.

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.5 Answer

Apply thin lens equation twice and multiply magnifications.